Early-Warning Fall Prediction for Assistive Robotics

A cost-sensitive, time-to-impact trade-off analysis on UMAFall

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The problem: a fall alarm after the fall is too late

- Falls are a leading cause of injury-related death and hospitalisation after age 65, and a safety risk for humanoid and assistive robots.
- **Detection** reports a fall when the person is already on the floor.
- **Prediction** must act in the few hundred milliseconds between losing balance and impact, when an airbag, a robot gait correction or an alert can still help.

My question

How early can a wearable-sensor classifier warn, and what does extra lead time cost?

- Lead time τ before impact
- Missed falls
- False alarms

What is new in this project

1. **Time-to-impact formulation:** the target is relabelled for every lead time τ (positive, negative and ambiguous zones), with strictly causal windows, so detection ($\tau = 0$) and early warning ($\tau > 0$) share one framework.
2. **Leakage-safe cost-sensitive alarm:** the threshold minimises $C_{FN} \cdot FN + C_{FP} \cdot FP$ on training subjects only (nested), with a sweep of the cost ratio.
3. **Event-level evaluation:** falls caught early, warning time and false alarms per minute, not just accuracy.
4. **Metric audit and decision layer:** a *strict* trial metric and temporal persistence of the score, reported with their limited benefit.
5. **Live streaming demo** that replays any trial window by window.

Not claimed

No new classifier: the models are standard; the contribution is the problem formulation, evaluation protocol and decision analysis.

Dataset: UMAFall (Universidad de Málaga)

Source: "UMAFall: Fall Detection Dataset", public on figshare, doi 10.6084/m9.figshare.4214283.

746	208	538	19
trials (15 s each)	simulated falls	daily activities	volunteers

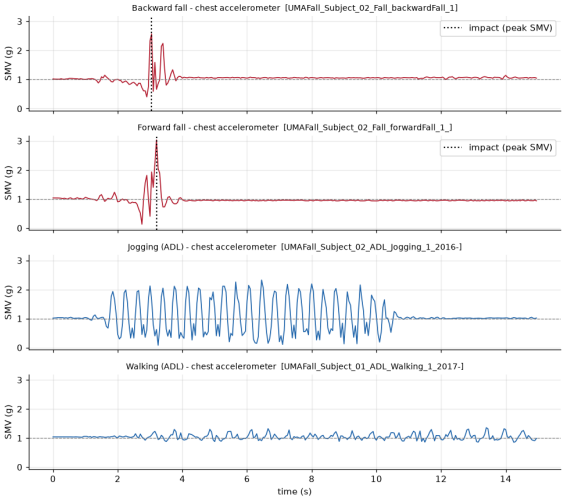
Sensors

- 5 positions: chest, waist, wrist, ankle, smartphone in right pocket
- Tags: accelerometer + gyroscope (+ magnetometer), ≈20 Hz
- Phone: accelerometer only, ≈200 Hz

Activities

- Falls: forward (71), backward (73), lateral (64)
- ADLs: walking, jogging, hopping, bending, sitting, lying down, stairs, clapping, door, phone call, hands up
- Vigorous ADLs are deliberate *hard negatives*

What a fall looks like in the signal

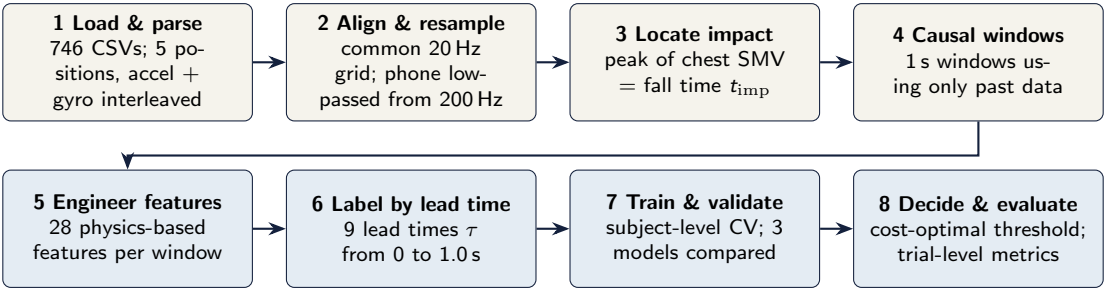


1. **Rest:** SMV $\approx 1\,\mathrm{g}$ (body supported)
2. **Free-fall dip:** support vanishes, SMV $\rightarrow 0\,\mathrm{g}$ *before* impact: the early-warning cue
3. **Impact spike:** sharp peak at the ground
4. **Stillness:** flat signal, too late to warn

$\mathrm{SMV} = \sqrt{x^2 + y^2 + z^2}$ is orientation-invariant.

Jogging reaches peaks as high as a fall.

Workflow: from raw CSV files to a cost-aware alarm



Preprocessing: data problems found, and what I did

Found in the data	**What I did**
Phone logs accelerometer only at ≈200 Hz; tags log 3 sensors at ≈20 Hz	Block-averaged and resampled the phone to 20 Hz; magnetometer left out
TimeStamp repeats for 2–3 rows (coarse clock)	Fitted a line of timestamp vs. sample index: evenly spaced time axis
Sensors missing in some trials (waist 617, ankle 643, chest 745 of 746)	Never assume 5 sensors; chest is the main position, others compared on the trials that have them
4 fall trials peak below 1.5 g; impact time ranges 1.9–14.6 s	Impact = peak SMV; all trials kept, listed as a limitation

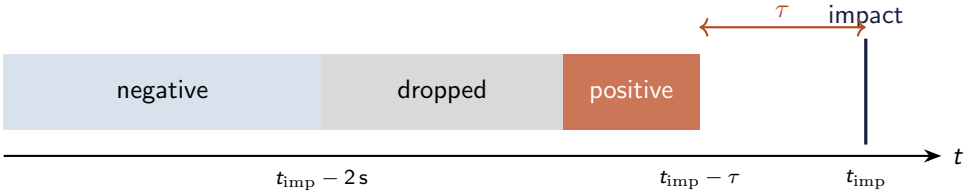
Feature engineering: 28 features tied to the physics

Group	Examples	Why
Free-fall dip	min SMV, dip depth, share of samples below 0.7 g	support vanishes before impact
Dynamics	mean, std, max, range, slope, jerk	sudden change
Orientation	tilt change (gravity direction, start to end)	body tips over
Rotation (gyro)	mean, max, std, slope of angular speed	loss of balance
Repetitiveness	2 s std, peak count, mean deviation	jogging vs. one event

Causal only

No post-impact stillness feature: it lies in the future, and early warning may only use data up to now.

The early-warning task: will impact happen about τ seconds from now?

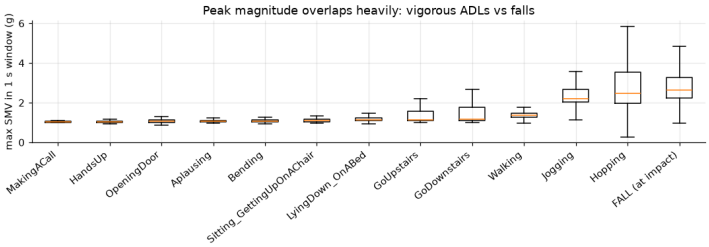


$\tau = 0$: window ends on impact, classic **detection**.

$\tau > 0$: only the dip and tilt before impact are seen: **early warning**.

9 lead times (0 to 1.0 s); about 1 456 positive vs. 78 545 negative windows (1 : 54).

Visualisation: why a simple threshold fails



Jogging and hopping reach peaks as high as a fall: peak size alone cannot separate them.

Single rule (ROC-AUC)	$\tau = 0$	$\tau = 0.3\,\mathrm{s}$
Peak SMV	0.652	0.479
Free-fall dip	0.826	0.759
Tilt change	0.868	0.771

Models and why

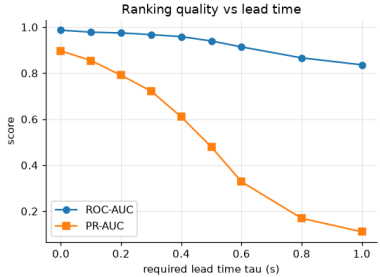
Three classifiers

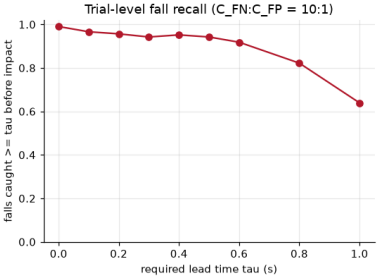
- **Logistic regression**: linear, interpretable baseline
- **Random forest (main)**: non-linear, robust to noisy features, copes with imbalance; lowest cost at $\tau = 0.3\,\mathrm{s}$
- **Gradient boosting**: strong tabular learner for comparison

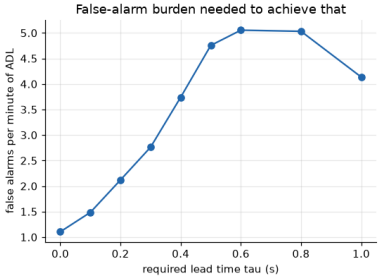
Validation and decision rule

- **Subject-level 5-fold CV**: nobody is in both train and test, so no memorising of people
- **Cost-sensitive threshold**: a missed fall costs $10\times$ a false alarm; chosen on training folds only (nested), then applied to unseen people
- Imbalance 1:54: class weights, PR-AUC, recall, false alarms per minute (never plain accuracy)

Results: more lead time makes the problem harder

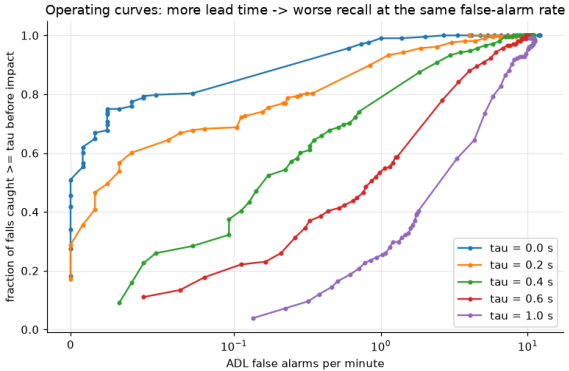






	$\tau = 0$	$\tau = 0.3\,\mathrm{s}$	$\tau = 0.6\,\mathrm{s}$
PR-AUC	0.90	0.72	0.33
Falls caught (trial level)	0.99	0.94	0.92
False alarms per minute	1.1	2.8	5.1

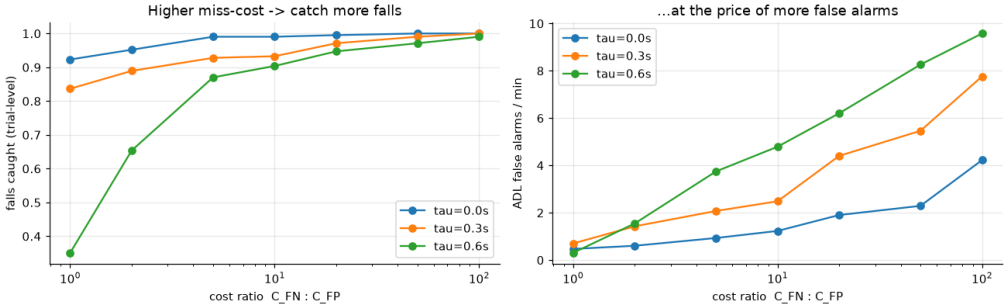
Results: the trade-off between recall and false alarms



τ (s)	PR-AUC	Falls caught	Alarms/min
0.0	0.90	0.99	1.1
0.2	0.79	0.96	2.1
0.4	0.61	0.95	3.7
0.6	0.33	0.92	5.1
0.8	0.17	0.82	5.0
1.0	0.11	0.64	4.1

Beyond $\approx$0.6 s even a generous false-alarm budget cannot recover recall.

Results: the cost ratio sets the operating point



- At $\tau = 0.3\,\mathrm{s}$: ratio 1:1 catches 84% of falls at 0.7 alarms/min; 20:1 catches 97% at 4.4; 100:1 catches 100% at 7.8.
- Choose the ratio by application: a hip airbag tolerates false alarms, a robot that stops walking does not.

Extension: strict evaluation and temporal persistence

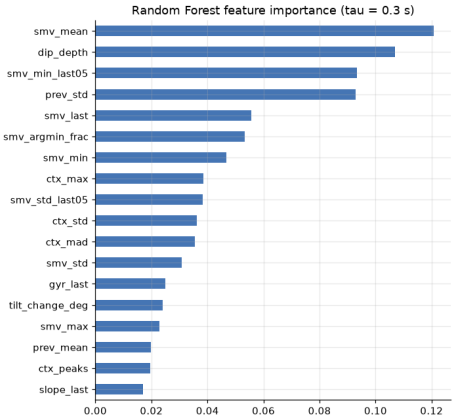
τ	Δ	Strict recall	Alarms/min	ADL w/ alarm
0.0	0	0.971	1.21	0.277
	0.4	0.962	1.21	0.272
0.3	0	0.913	2.46	0.508
	0.2	0.899	2.41	0.490
	0.4	0.899	2.48	0.501
0.6	0	0.894	4.91	0.814
	0.2	0.846	4.32	0.771
	0.4	0.817	3.72	0.717

- **Strict metric:** only alarms in $[t_{\mathrm{imp}} - 2\,\mathrm{s},\ t_{\mathrm{imp}} - \tau]$ count; early alarms in fall trials are false alarms.
- It lowers recall by only **2–3 points**: the headline result is robust.
- **Persistence** (Δ = mean score over last Δ s): no effect at 0 and 0.3 s; at 0.6 s alarms -24% but recall -7.7 points.
- A trade-off, not a clear gain: a negative result for short horizons.

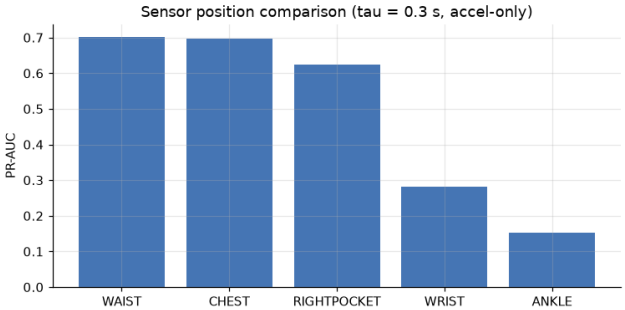
Model comparison and feature importance ($\tau = 0.3\,\mathrm{s}$)

Model	ROC	PR	Cost	Alarms/min
Logistic reg.	0.941	0.472	5714	3.70
Random forest	0.968	0.722	4156	2.76
Grad. boosting	0.960	0.710	4454	3.95

Top features: window mean SMV, dip depth, min SMV in the last 0.5 s, std of the previous second: a drop below the preceding level after a calm period.



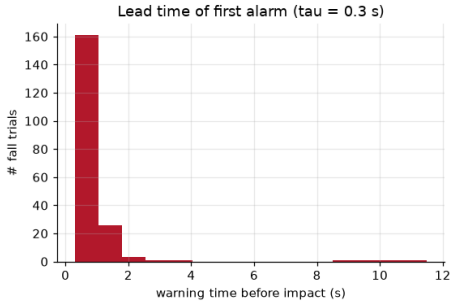
Which sensor position works best? ($\tau = 0.3\,\mathrm{s}$, accelerometer only)

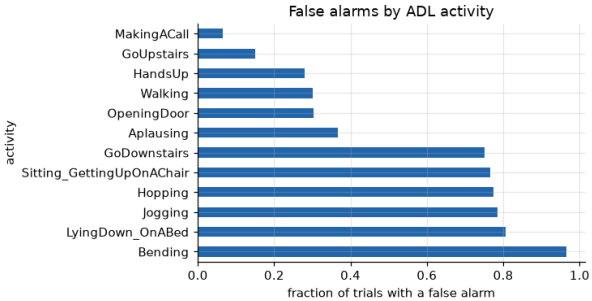


Position	PR-AUC	Alarms/min
Waist	0.701	3.18
Chest	0.698	2.85
Pocket	0.623	3.74
Wrist	0.282	6.05
Ankle	0.152	4.60

- Chest and waist (near the centre of mass) are best; a phone in the pocket loses little.
- Wrist and ankle lose most of the signal: limb motion is dominated by other activities.

Error analysis ($\tau = 0.3\,\mathrm{s}$)





- Falls caught by type: backward 96%, forward 94%, lateral 92%.
- Most false alarms: bending (97% of trials), lying down (81%), jogging (79%), hopping (77%), sitting/getting up (77%).
- Fewest: making a call (7%), going upstairs (15%), hands up (28%), walking (30%).

Limitations and honest caveats

- Falls are simulated by young, healthy volunteers; elderly falls differ.
- Only 19 subjects; one subject has 56 of the 208 falls; single seed, no confidence intervals.
- 20 Hz sampling: the dip is resolved only to 50 ms.
- Impact = peak SMV: wrong for a few suspicious trials.
- False-alarm burden is high: at $\tau = 0.3$ s, 55% of ADL trials raise at least one alarm.
- Trial-level recall counts any alarm before $t_{\mathrm{imp}} - \tau$ (0.94 vs. window-level 0.80 at $\tau = 0.3$ s); the strict metric lowers it by only 2–3 points.

Conclusions and future work

1. Falls leave an early physical signature (free-fall dip and tilt) that a classifier can use before impact.
2. Early warning is a trade-off: at $\tau = 0.6\,\mathrm{s}$ PR-AUC falls from 0.90 to 0.33 and false alarms rise from 1.1 to 5.1 per minute.
3. The cost ratio lets the application choose its operating point; chest and waist sensors work best, wrist and ankle lose much.
4. Strict evaluation changes recall by only 2–3 points; temporal persistence reduces false alarms only at 0.6 s, and at a cost in recall.

Future work

Orientation-corrected features (magnetometer), sequence models (1-D CNN / LSTM), fusion of several positions, personalised thresholds, other datasets (e.g. SisFall) and data from older adults.

Thank you